

DS Semester Examination 2025

Data Structures (DS) · Semester Examination · 2025

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Compare Asymptotic Notation with Singly Linked Lists. When would you prefer one over the other? (5 marks)
2. Define Stack ADT. Derive or justify the key properties involved. (5 marks)
3. What is meant by Binary Tree Traversals? Discuss its advantages and limitations. (5 marks)
4. What is meant by Graph Representations? Discuss its advantages and limitations. (5 marks)
5. Write short notes on Binary Search. (5 marks)

Part B

1. Define Asymptotic Notation. Derive or justify the key properties involved. (Unit 1)
2. Illustrate Arrays with a worked example. (Unit 1)
3. Illustrate Stack ADT with a worked example. (Unit 2)
4. Discuss the role of Infix to Postfix Conversion in Stacks & Queues. (Unit 2)
5. Define Binary Tree Traversals. Derive or justify the key properties involved. (Unit 3)
6. Define Binary Search Trees. Derive or justify the key properties involved. (Unit 3)
7. Compare Graph Representations with Depth-First Search. When would you prefer one over the other? (Unit 4)
8. Illustrate Breadth-First Search with a worked example. (Unit 4)
9. Compare Binary Search with Hash Functions. When would you prefer one over the other? (Unit 5)
10. Define Quick Sort. Derive or justify the key properties involved. (Unit 5)